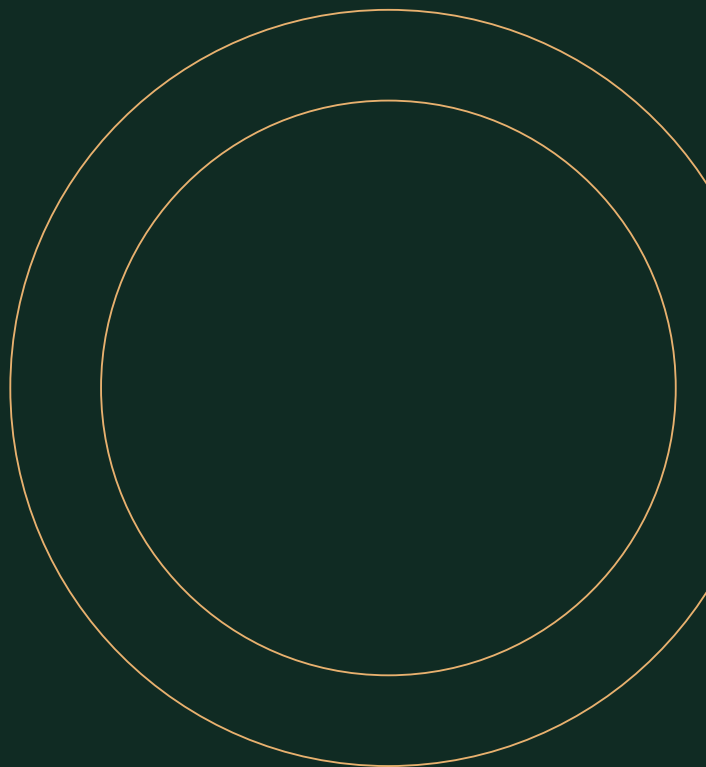


Evidence controlled.

Intelligence governed.

End-to-end product, system, data, AI, operations and deployment specification for governed commissioning intelligence.



DOCUMENT INDEX

A4 technical dossier

This print-safe edition consolidates the current repository architecture, product requirements, technical requirements, data schema, application flow, design decisions, retrieval safeguards, release ledger and browser acceptance evidence.

Part	Content
01	Purpose, target users and authority model
02	Application experience and live screenshots
03	System architecture and service boundaries
04	End-to-end data flow, user flow and navigation
05	Database, provenance and schema domains
06	AI, RAG, agents and safety controls
07	Technology stack, SDKs, APIs and third parties
08	Prerequisites, installation, environment and operations
09	Testing strategy, branch/application errors and mitigations
10	Deployment readiness and future CI/CD

01 / PURPOSE, USERS AND AUTHORITY

What Pramana Cx is designed to decide

Pramana Cx is a governed evidence control plane for mission-critical EPC and data-centre commissioning. It connects controlled requirements to systems, assets, gates, tests, evidence, findings, schedules, shipments and immutable source evidence so an authorized engineer can make a defensible gate decision.

Non-negotiable authority rule. AI may extract, retrieve, rank, draft, classify and explain. Deterministic services calculate readiness, schedule feasibility and structured verdicts. Only an authorized human can accept a requirement or evidence item, approve a report, disposition a compliance finding or make a gate decision.

Target users and friction removed

User	Decision	How the application makes it easier
Commissioning manager / authority	Can a gate be defended today?	Readiness unifies accepted proof, stale/failed evidence, predecessors, blockers, owners and citations.
EPC / GC MEP package lead	What must be fixed before the next test?	Actions, Cx, evidence and schedule views preserve shared operational context.
QA/QC engineer	Does a controlled record meet a requirement?	Source-region provenance, deterministic checks and cited comparison proposals replace evidence hunting.
Planner / scheduler	What changed and should the baseline move?	Reviewed inputs, immutable CP-SAT versions and before/after explanations separate facts from speculation.
Owner / operations team	Is turnover complete and auditable?	Approved-gate-only manifests, hashes and independent verification make the package inspectable.
Field engineer / vendor	How can proof be captured safely on site?	Offline capture queues evidence idempotently until an authorized reviewer accepts it.

Application coverage

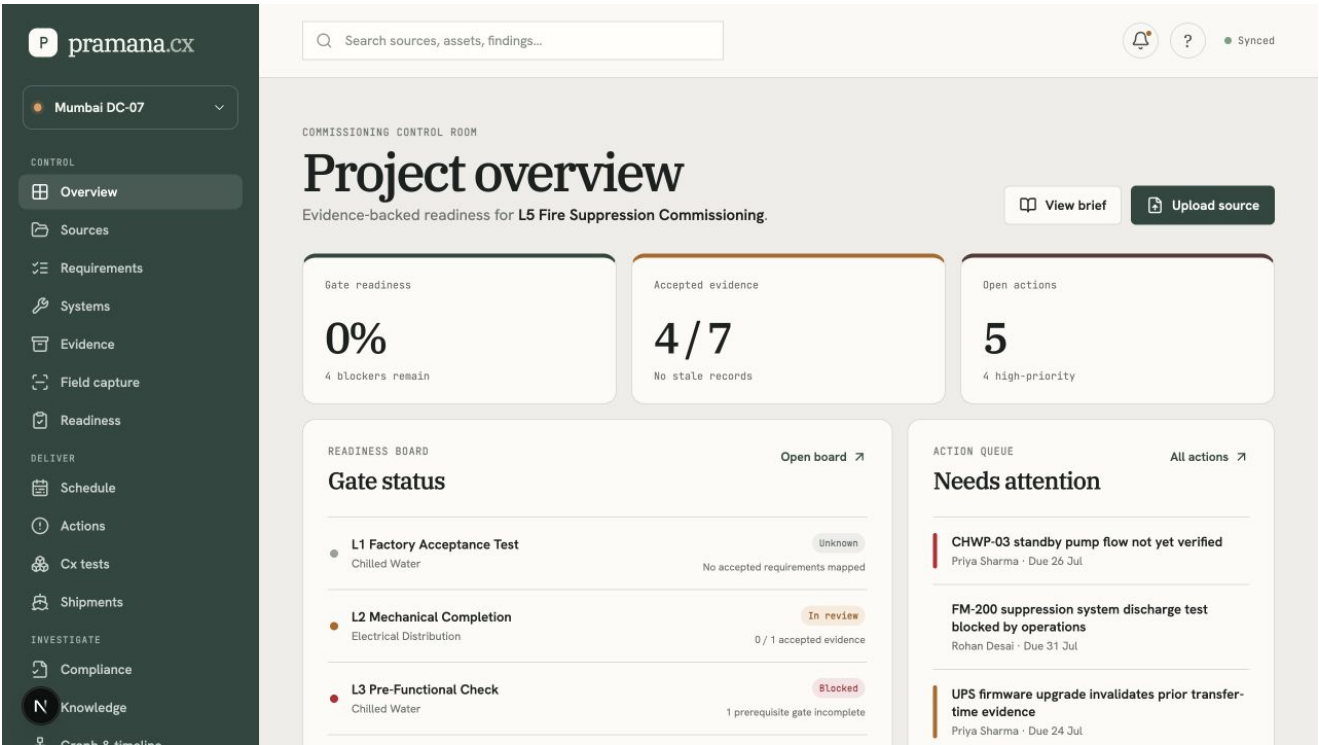
- Controlled sources, revisions, exact source-region citations and proposal review.
- Requirements, systems, assets, gates, evidence, findings, readiness and turnover packs.
- Schedule inputs, CP-SAT baselines, immutable versions, live events and advisory risks.
- Cx standards, cited checklists, deterministic readings, reports and human approval.
- Shipments, map routes, compliance, knowledge/RFI intelligence, graph, changes and Command Center.

02 / APPLICATION EXPERIENCE

Task-oriented, cited and resilient

The workspace groups navigation as Control, Deliver and Investigate. Deep links preserve the selected gate, task, finding or shipment across feature boundaries. Busy-state and progress behaviour is operation-specific, bounded and visible, so long-running work cannot silently replace another action.

Live browser-captured application surfaces



Overview and grouped navigation - one project context and a complete task map.

pramana.cx

Mumbai DC-07

CONTROL

Overview

Sources

Requirements

Systems

Evidence

Field capture

Readiness

DELIVER

Schedule

Actions

Cx tests

Shipments

INVESTIGATE

Compliance

Knowledge

Graph & timeline

Search sources, assets, findings...

Online · server connected

COMMISSIONING QA COPILOT

Cx tests

Ingest controlled standards, generate citation-verified advisory checklists, execute deterministic proposed checks, and approve an editable report into immutable evidence.

Overview

1 · CONTROLLED CORPUS

Ingest standards/procedures

Checklist generation remains disabled until extraction creates exact citation regions.

STANDARD SET

ASHRAE / project IST set

TITLE

Synthetic chilled-water IS

DOCUMENT TYPE

Standard

REVISION

Rev 1

PDF

Choose file

No file chosen

Control & extract

2 · ADVISORY GENERATION

Generate cited draft

5 extracted standard version(s) available.

TITLE

L4 integrated systems test

SYSTEM

Select system

GATE

Select gate

EQUIPMENT

Select equipment

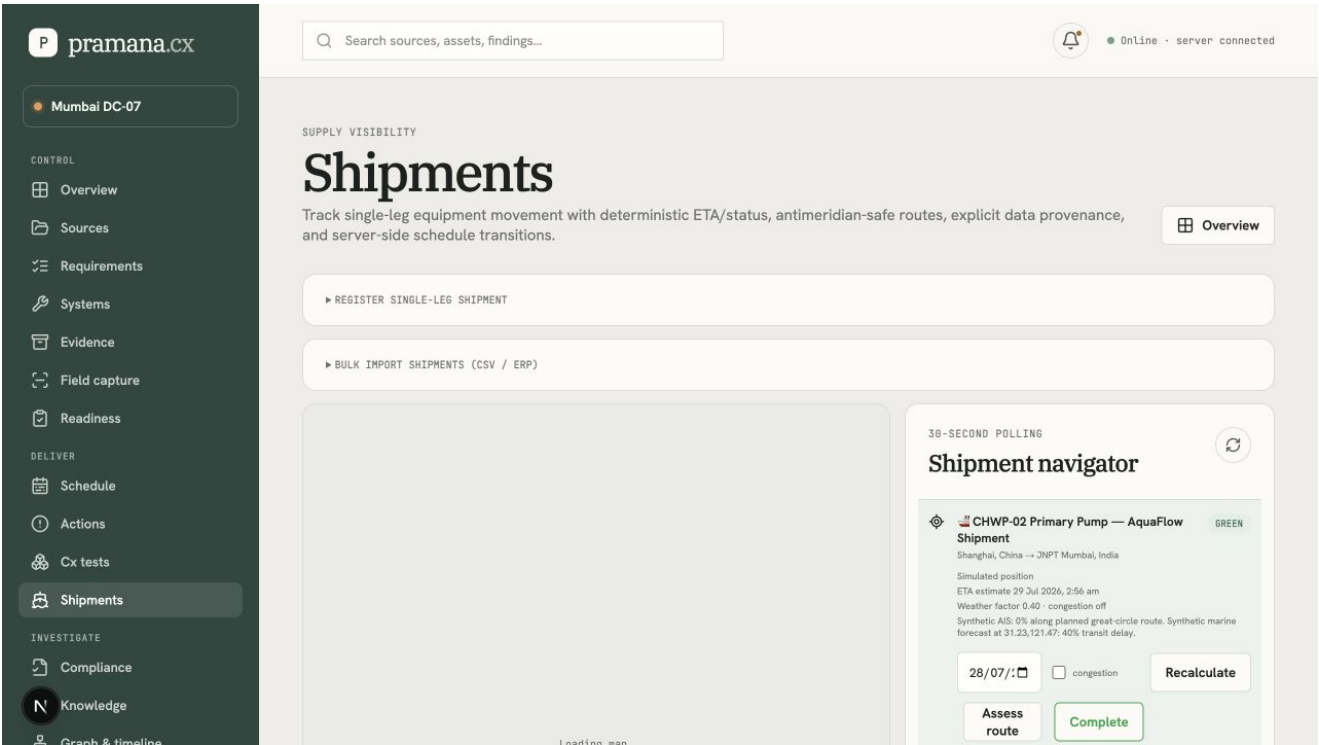
GOVERNING STANDARD VERSIONS

TIA-942 · tia · 1
Unclassified set · CHW Plant Commissioning Procedure · Rev C
ASHRAE · ASHRAE Std 90.1-2022 — Energy Standard · Rev 2022
NFPA · NFPA 2001 — Clean Agent Fire Suppression · Rev 2018
Unclassified set · CHW Plant Commissioning Procedure · Rev C

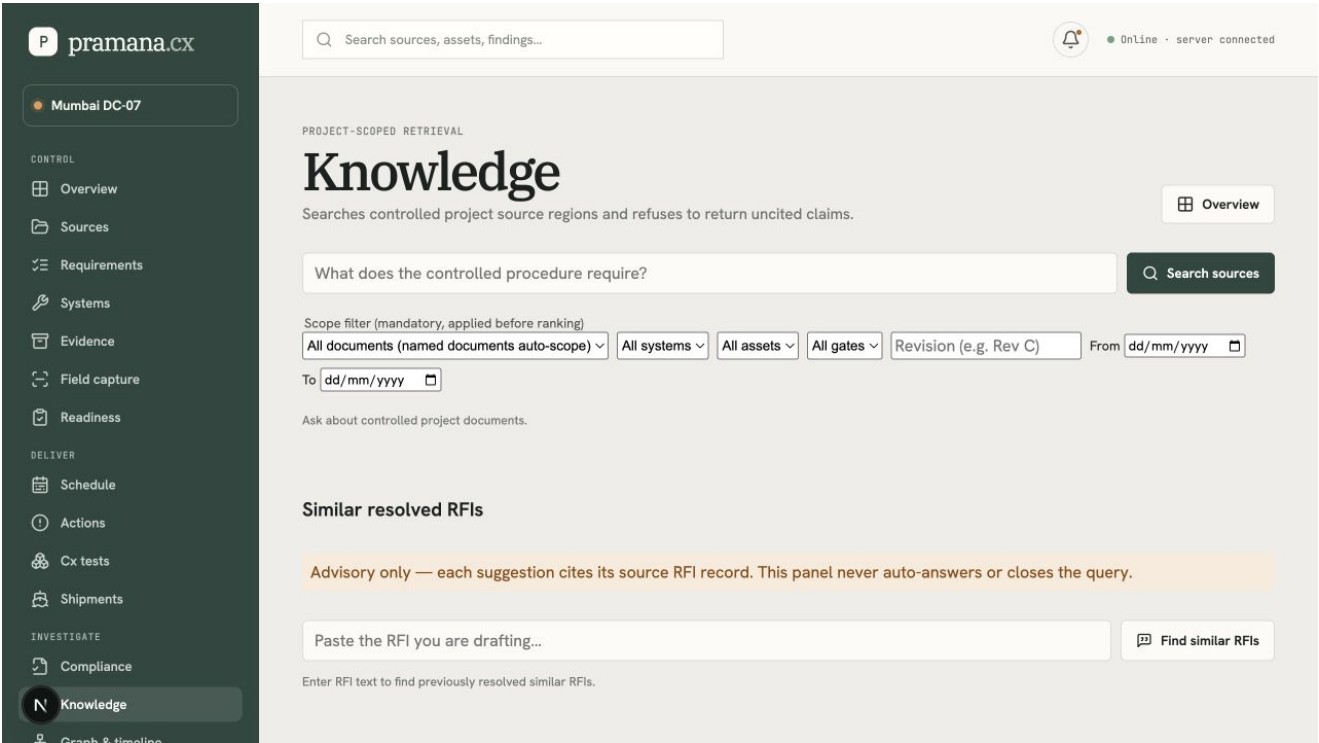
Generate draft

Governed Cx tests - cited standards, deterministic readings, human review and report approval.

CONFIDENTIAL | PAGE 5



Shipment visibility - route/position context, ETA provenance and controlled operational status.



Cited knowledge intelligence - project-scoped retrieval and source-grounded answers.

End-to-end user journey

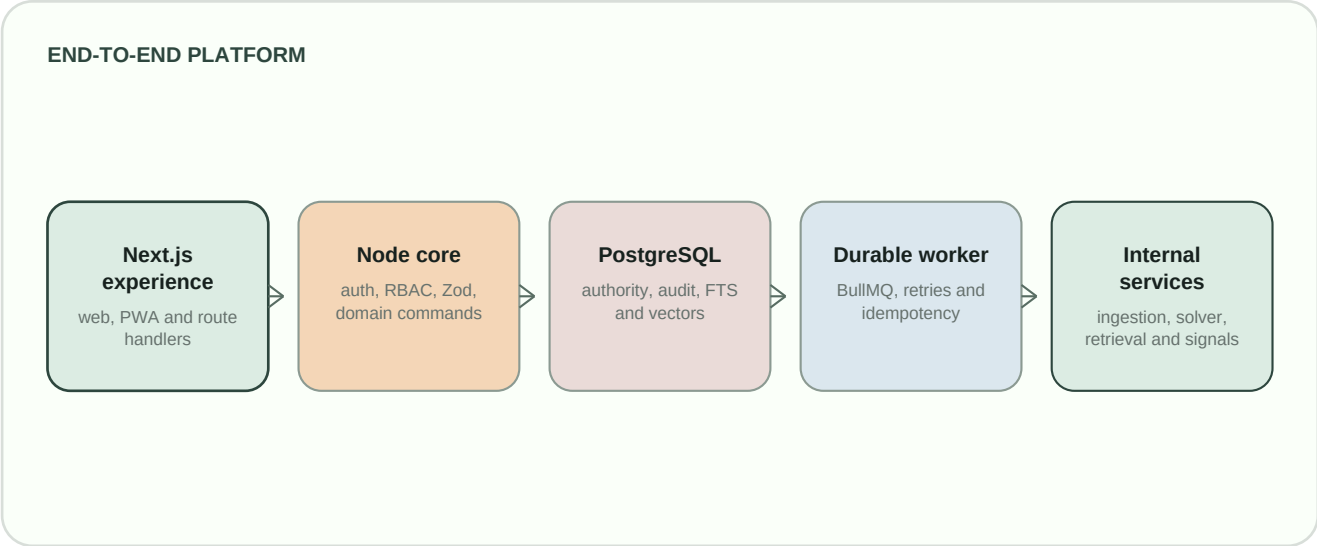
Step	Surface	Data flow	Outcome
1	Overview / Settings	Project membership scopes all loads and actions.	User enters an authorized project context.
2	Sources / Cx	Bytes -> hash/version -> source regions -> durable extraction.	Reviewer sees cited proposals.
3	Systems / Requirements	Accepted records create typed system/asset/gate relationships.	Facility context is traversable.

Step	Surface	Data flow	Outcome
4	Evidence / Field / Cx	Evidence starts pending; review adds acceptance/audit state.	Proof contributes to readiness only when accepted.
5	Schedule / Shipments	Validated event -> delta detector -> CP-SAT only if material.	New immutable version or status-only update.
6	Actions / Compliance / Knowledge	Findings, alerts and citations retain record identity.	Owner resolves human-reviewed work.
7	Readiness / Turnover	Rules + authorized decision -> manifest.	Approved gates produce verifiable turnover.

03 / SYSTEM ARCHITECTURE

Modular monolith with isolated services

The Next.js core is the authenticated, project-scoped front door and business authority. Durable or expensive work passes through Redis/BullMQ. PostgreSQL is the transactional source of truth; object storage holds immutable bytes; typed relational edges provide governed graph traversal.



Service ownership and boundaries

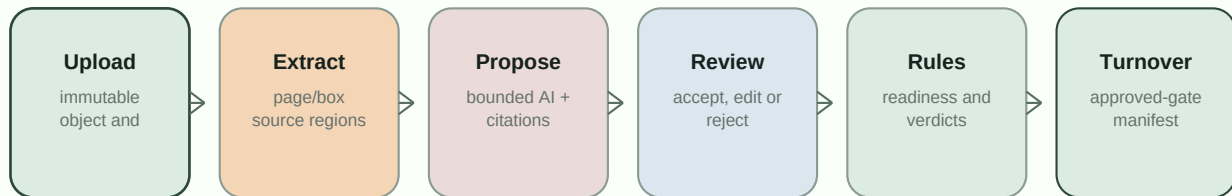
Component	Owns	Must not own
Next.js core	Authentication, authorization, validation, domain commands, audit and project scoping.	Long solver/extraction work or vendor-specific direct AI calls.
Worker + BullMQ	Retryable background operations, stable job identity and durable job state.	Interactive authorization or final human approval.
Ingestion service	Format extraction into structured regions.	Business authority, readiness or permission decisions.
CP-SAT solver	Feasible schedule or explicit bottleneck/infeasibility result.	Database/object storage access, approvals or LLM calls.
Retrieval service	Embeddings and cross-encoder reranking.	Database credentials, tenant decisions or final-answer authority.

Architecture decision. CP-SAT runs as a stateless dedicated service because an optimization run must not block the Node request process. The caller passes the complete DAG/constraints; the solver does not read Postgres or object storage.

04 / DATA FLOW, USER FLOW AND NAVIGATION

From controlled source to approved turnover

AUTHORITY AND EVIDENCE FLOW

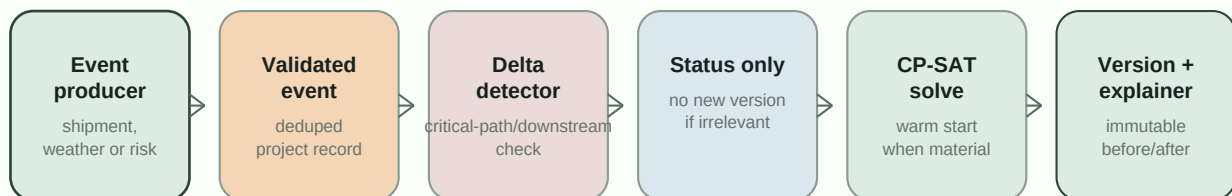


Runtime job lifecycle

- The browser submits a project-scoped command. The core authenticates, authorizes, validates and persists idempotency/audit state.
- Long-running work becomes a stable BullMQ job. The caller receives a job reference and can read progress without freezing the UI.
- The worker reads immutable input, invokes an internal service under timeout/retry rules, validates structured output and persists result, edges and audit event.
- The UI reads the authoritative result and labels advisory versus authoritative state.

Schedule event flow

MATERIALITY-GATED SCHEDULE HANDLING



Schedule safety boundary. Predictive risk can observe, explain and propose mitigation options. It cannot apply a mitigation, alter dependencies/resources, invoke the solver autonomously or change dates.

Navigation contract

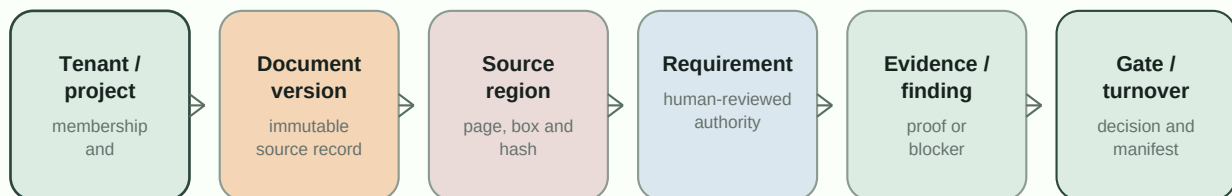
Cross-feature actions include a stable identifier in the URL, for example `/readiness?gate=...`, `/schedule?task=...`, `/actions?finding=...` and `/shipments?shipment=...`. The destination validates, consumes and focuses the exact record; a missing target degrades safely rather than producing an unrelated generic screen.

05 / DATABASE, PROVENANCE AND SCHEMA

Transactional authority, immutable bytes, traversable relationships

Every authoritative record is tenant and project scoped. Cross-project identifiers are rejected at the API boundary. Normalized relational tables remain the authority; the edges relation enables governed traversal; immutable object storage retains original source/evidence/report/turnover bytes.

CORE ENTITY PATH



Persistence domains

Domain	Representative records	Integrity intent
Identity and scope	tenants, users, projects, project_members, sessions	Membership is evaluated server-side for every record and action.
Sources and knowledge	documents, document_versions, source_regions, knowledge_chunks	Immutable bytes, exact citations and embedding-model tags prevent mixed vector spaces.
Evidence control	requirements, evidence, systems, assets, gates, findings, decisions, turnover packs	Human review and deterministic readiness govern state transitions.
Planning and supply	schedule_tasks, resources, schedule_versions, schedule_events, risks, shipments	Accepted inputs, immutable solve history, event deduplication and provenance.
Cross-cutting	edges, durable_jobs, alerts, audit_events	Typed relationships, retry/idempotency and append-only hash-linked history.

Provenance safeguards

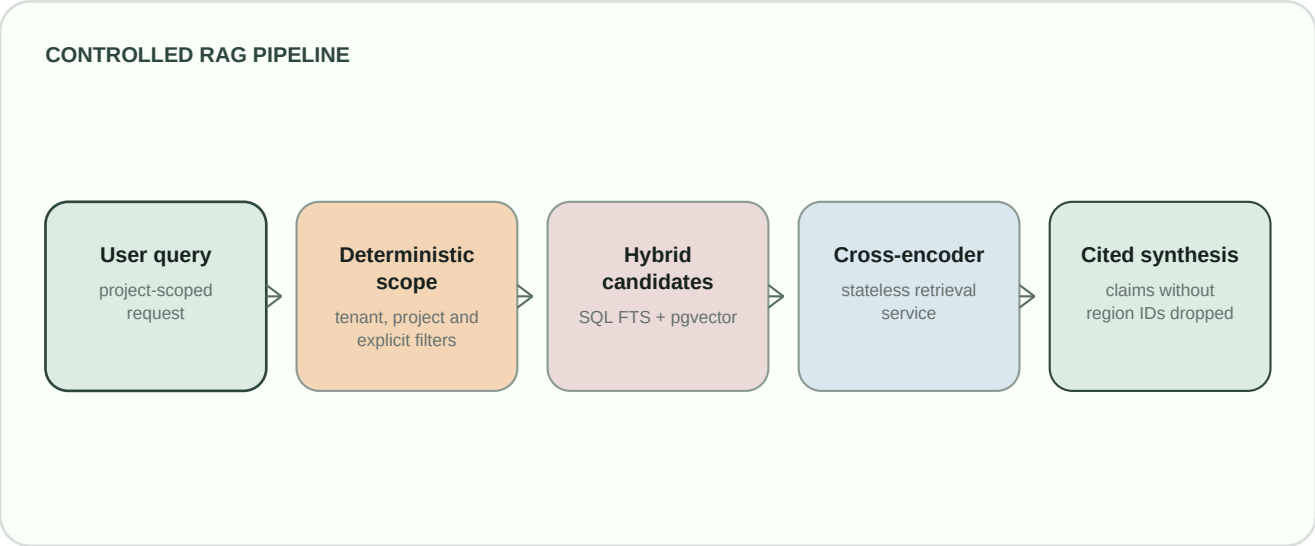
- Immutable SHA-256 object record is stored before extraction; source regions retain document version, page, bounding box and content hash.
- New revisions never overwrite old regions; superseded, rejected, failed or unauthorized sources cannot be retrieval candidates.
- Accepted-only evidence can prove a requirement. Stale/failed evidence invalidates readiness; edits trigger audited recalculation.

- Audit events are append-only and hash-linked per project.

06 / AI, RAG AND AGENT BOUNDARIES

Bounded, cited, time-limited and non-authoritative

Core generation and embedding calls resolve through a central provider boundary. Local Ollama is the intended deployment default: gemma4:e2b for structured generation and nomic-embed-text:latest for 768-dimensional embeddings. Gemini and NVIDIA NIM can be selected explicitly. The deterministic mock provider is verification-only and must never be deployed.



Provider and RAG safeguards

- Bounded prompt/context sizes, output token caps, request deadlines, schema validation and retry limits.
- Client aborts, per-operation busy state and progress feedback prevent frontend blocking.
- Tenant/project and supplied metadata filters are enforced in SQL before ranking, not after retrieval.
- Only explicit reviewer/user filters and deterministic title routing may narrow authority-bearing scope. The LLM cannot choose system, asset, gate, revision, type or document scope.
- Groundedness is code-enforced: an answer claim without a resolvable source-region identifier is dropped.

Five operational agent capabilities

Capability	Implementation posture	Non-negotiable boundary
Commissioning QA Copilot	Cited standards, checklist proposal, deterministic readings, human-routed narrative review and draft report workflow.	No automatic approval; approved report only becomes immutable evidence after review.
Supply Chain Visibility and Risk	Shipment legs, multi-mode map, polling and weather/congestion/AIS assessment.	Delayed/recovered event is provenance-labelled and cannot directly alter a schedule.
Specification and Quality Compliance	Semantic candidate discovery plus deterministic numeric/boolean/category comparisons and precedent handling.	Creates proposed findings only; an engineer decides disposition.
Predictive Schedule Risk	Bounded observations of procurement, lead-time, workforce and weather with materiality/deduplication.	Advisory risks/mitigations only; it cannot change dates or apply an option.
Knowledge and RFI Intelligence	Hybrid project-scoped retrieval, graph context, reranking, cited synthesis and similar-RFI retrieval.	Returns grounded citations or explicit no-results; never invents authority.

Concrete technology inventory

Layer	Technology	Purpose
Web	Next.js 16.2.10, React 19, TypeScript 5.7	Server-rendered application, route handlers, typed UI and responsive PWA surfaces.
Schema and validation	Drizzle ORM 0.45, Drizzle Kit 0.31, Zod 3.24	SQL migrations, relational authority and validated contracts.
Database	PostgreSQL 16 target topology, pgvector, Postgres FTS	Transactional state, constraints, audit/jobs and hybrid search.
Queue/cache	Redis, BullMQ 5.80, ioredis 5.11	Durable background work, idempotency coordination and rate limits.
Storage	Filesystem locally; MinIO/S3 via AWS SDK v3	Immutable original, report and turnover artifacts with signed reads.
Extraction / solver / retrieval	Python/FastAPI, PyMuPDF, Google OR-Tools CP-SAT, BAAI bge models	Controlled source extraction, optimization, embeddings and reranking.
AI providers	Ollama gemma4:e2b and nomic-embed-text; optional Gemini and NVIDIA NIM	Explicit schema-validated advisory provider choices.
Maps and UX	Leaflet, React Leaflet, Turf, searoute-js, OpenStreetMap, Lucide, Three.js	Route visualisation, geospatial geometry and application visual system.
Identity	bcryptjs, OTPAuth, optional Clerk Next.js SDK	Owned credentials/TOTP or external identity adapter with app-owned RBAC.

API group contract

Group	Representative routes	Expectation
Identity	/api/auth/*, /api/profile	Credentials/Clerk, sessions, TOTP and rate limits.
Project authority	/api/projects, /members, /activate, /alerts	Server-side membership and role enforcement.
Evidence control	/sources, /evidence, /requirements, /gates, /turnover-packs	Versioning, citations, review and readiness.
Schedule and risk	/schedule/tasks, /resources, /baseline, /events, /risks	Reviewed input, immutable versions and solver isolation.
Cx/compliance/knowledge	/cx/*, /compliance/*, /knowledge/*, /graph	Cited, scoped, proposed-only advisory flows.

08 / PREREQUISITES, INSTALLATION AND CONFIGURATION

Local topology and production configuration

Local development can use separate PostgreSQL, Redis, ingestion and solver processes or the complete Compose topology. Production must use secret-managed configuration, non-degraded infrastructure and a model endpoint reachable from both web and worker services.

Installation

```
npm install
cp .env.example .env.local
npm run db:migrate
npm run db:seed
npm run system:start

npm run typecheck
npm run build
npm run verify:all

ollama pull gemma4:e2b
ollama pull nomic-embed-text:latest
MODEL_PROVIDER=ollama EMBEDDING_PROVIDER=ollama npm run verify:ollama
```

Environment and infrastructure requirements

Requirement	Local	Production
Runtime	Node 22, npm, Python 3; Docker Desktop for Compose.	Pinned images/runtimes; separate web and worker processes.
Database	Reachable Postgres; migrations applied; pgvector where used.	TLS, backups, restore rehearsal, role separation and empty-db migration proof.
Secrets	.env.local never committed.	Secret manager, rotation policy and no secrets in artifacts/logs.
Object storage	Filesystem or local MinIO.	Private bucket, encryption, lifecycle/retention and signed URL policy.
Models	Optional local Ollama for real smoke tests.	Reachable model service for web/worker with data-processing approval.

Key configuration groups

Group	Representative keys	Release condition
Auth	APP_BASE_URL, AUTH_MODE, AUTH_ENCRYPTION_KEY, Clerk keys	Use credentials or Clerk, never development mode.
AI/RAG	MODEL_PROVIDER, EMBEDDING_PROVIDER, OLLAMA_BASE_URL, model keys, GEMINI_API_KEY, NIM_API_KEY	No mock provider; endpoints reachable and bounded.
Infrastructure	Database/Redis URLs, object storage settings, INFRA_ALLOW_DEGRADED	All dependencies healthy; degraded mode false.
Signals	Procurement, lead-time, workforce, weather, AIS/congestion settings	Live/synthetic/unavailable provenance is explicit.

Installation exit criteria. Migrations pass, dependencies are healthy, an authorized source can be processed, the verification matrix passes and the real-provider smoke test is run separately when a provider is configured.

09 / TESTING, ERROR HISTORY AND MITIGATION

Release confidence is a matrix, not a green build

Area	Coverage	Evidence
Build/migration	TypeScript, optimized build, Drizzle checks and migration replay.	Valid artifact and schema state.
Security/tenancy	Credentials/TOTP, rate limits, cross-project rejection and audit chain.	Server-side enforcement and append-only history.
Evidence to turnover	Pending/accepted proof, readiness, gate decision and manifest verification.	Accepted-only proof and independently verifiable turnover.
Schedule/risk	Inputs, dependencies, CP-SAT feasibility, immutable versions and recovery.	No solve for irrelevant event; explicit degradation.
Cx/compliance/knowledge	Citations, deterministic readings, review, filters, reranking and groundedness.	No ungrounded or unauthorized decision.
Browser acceptance	Sidebar journeys, mobile/desktop layouts, busy state, maps and console capture.	No duplicate-key warning, error boundary or overflow.

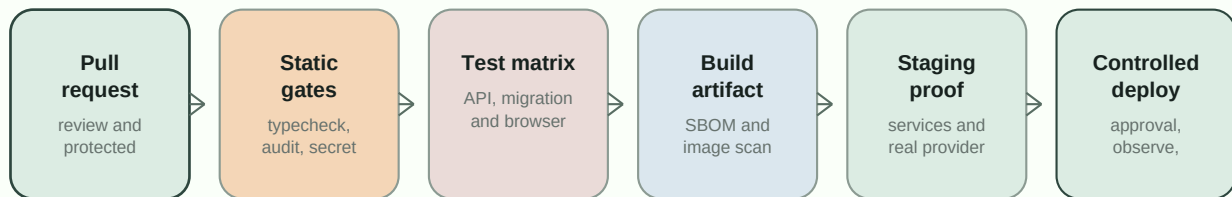
Cross-branch and application mitigation ledger

Failure found	Mitigation now built
Mock TypeScript output could impersonate an LLM.	Explicit generation/embedding providers, Ollama default, mock test-only rule and provenance.
Direct Gemini/provider bypasses produced inconsistent policy.	Central schema-validated provider boundary and real-provider smoke tests.
Long prompts made operations look stuck.	Prompt/token/request bounds, client abort, progress, busy state and bounded retry.
Deep links lost record context.	Destination validation and focus contracts for readiness, schedule, actions and shipments.
Duplicate React keys and business rows.	Stable composite UI keys, unique business constraints and integrity verification.
Conflicting branch migrations.	Final-schema-first reconciliation, ordered migrations and replay validation.
Processed documents were absent/misattributed in RAG.	Extraction-indexing contract, backfill, SQL document filtering and citation validation.
Map fallbacks looked operational.	Coordinate checks, HTTPS/deadline routing, mode-aware provenance and explicit fallback notice.
Raw JSON/hashes/synthetic cues degraded UX.	Presentation helpers, human labels, bounded formatting and source-state labels.
Architecture documentation regressed through merge work.	Visible README diagrams, this PDF dossier, release ledger and documentation-first review.

Local verification result. Type check, optimized production build, production dependency audit and the post-fix local verification matrix have passed. Target-environment infrastructure, real provider acceptance, accessibility and load validation remain production gates.

10 / DEPLOYMENT READINESS AND FUTURE CI/CD

Controlled deployment, not an environment-variable switch

FUTURE CI/CD PIPELINE**Required production gates**

- Secrets and identity: production AUTH_MODE, encryption/session settings, base URL and identity credentials are secret-managed and validated.
- AI: selected Ollama, Gemini or NIM generation/embedding endpoint is reachable by web and worker; deadline, structured output and embeddings are smoke-tested in release configuration.
- Data plane: pgvector is available; migrations-from-empty, S3/MinIO lifecycle, Redis/worker recovery and backup/restore are rehearsed.
- Internal services: ingestion, solver and retrieval health checks are green; TLS/internal network restrictions and non-degraded infrastructure are active.
- Live signals: AIS/weather/congestion/procurement provenance and failure semantics are validated; synthetic paths are disabled or visibly labelled outside test/demo environments.
- Operations: CI, structured logs/correlation IDs, metrics, traces, queue dashboards, alerting, incident ownership, accessibility and load testing are present before release.

Explicit non-claims

Engineering authority remains human. Pramana Cx assists engineering judgement. It does not independently certify a facility, replace the commissioning authority or engineer of record, issue Tier/TIA/BICSI/statutory approval, autonomously close findings/waivers or represent simulated operational feeds as live evidence.

Maintenance references

README.md - implementation map, diagrams, defect audit and configuration. **STATUS.md** - authoritative shipped/open ledger. **PLANNER/PRD.md** and **PLANNER/TRD.md** - product scope and technical contracts. **PLANNER/Schema.md**, **AppFlow.md**, **DesignDecisions.md** and **RetrievalArchitecture.md** - data model, flows, decisions and retrieval safeguards.